

Mac 1

princekumarsingh@Princes-MacBook-Pro-3 ~ % sudo lsof -nP -i :53

Password:

COMMAND	PID	USER	FD	TYPE	DEVICE	SIZE/OFF	NODE	NAME
dnsmasq	6915	nobody	4u	IPv4	0x2c152034f30e94f8	0t0	UDP	10.7.28.177:53
dnsmasq	6915	nobody	5u	IPv4	0x2d320aba5261b46d	0t0	TCP	10.7.28.177:53 (LISTEN)
dnsmasq	6915	nobody	6u	IPv4	0xfaeae424b1245c10	0t0	UDP	127.0.0.1:53
dnsmasq	6915	nobody	7u	IPv4	0xb8ea262947edfbef	0t0	TCP	127.0.0.1:53 (LISTEN)
dnsmasq	6915	nobody	8u	IPv6	0x97ae05800c8fc9b9	0t0	UDP	[fe80:b::10c8:c644:a563:e270]:53
dnsmasq	6915	nobody	9u	IPv6	0x3e79419f40f2d008	0t0	TCP	[fe80:b::10c8:c644:a563:e270]:53 (LISTEN)
dnsmasq	6915	nobody	10u	IPv6	0x8ab3671def79aa1c	0t0	UDP	[fe80:1::1]:53
dnsmasq	6915	nobody	11u	IPv6	0xe50e9ef8a109853d	0t0	TCP	[fe80:1::1]:53 (LISTEN)
dnsmasq	6915	nobody	12u	IPv6	0xc63b6a773bef0c48	0t0	UDP	:::1:53
dnsmasq	6915	nobody	13u	IPv6	0xbba85cdc8178e4f	0t0	TCP	:::1:53 (LISTEN)

princekumarsingh@Princes-MacBook-Pro-3 ~ %

princekumarsingh@Princes-MacBook-Pro-3 ~ % tail -20 /tmp/dnsmasq.log

tail: /tmp/dnsmasq.log: Permission denied

Mac 2

atanuadhikari@Atanus-MacBook-Pro ~ % dig @8.8.8.8 app.team1.test

; <<>> DiG 9.10.6 <<>> @8.8.8.8 app.team1.test

; (1 server found)

;; global options: +cmd

;; Got answer:

;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 43053

;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:

; EDNS: version: 0, flags::, udp: 512

;; QUESTION SECTION:

;app.team1.test. IN A

;; AUTHORITY SECTION:

. 85738 IN SOA a.root-servers.net. nstld.verisign-grs.com.

2026100300 1800 900 604800 86400

;; Query time: 10 msec

;; SERVER: 8.8.8.8#53(8.8.8.8)

;; WHEN: Sat Oct 03 11:32:06 IST 2026

;; MSG SIZE rcvd: 118

atanuadhikari@Atanus-MacBook-Pro ~ %

Mac 3

```
(base) sambhavkumar@Sambhavs-MacBook-Pro ~ % lsof -nP -iTCP -sTCP:LISTEN | grep -E ":3001|:3002"
python3.1 59654 sambhavkumar 3u IPv4 0x30d6eebf0f8907e5 0t0 TCP *:3002 (LISTEN)
python3.1 66937 sambhavkumar 3u IPv4 0x30d6eebf0fc09ed5 0t0 TCP *:3001 (LISTEN)
(base) sambhavkumar@Sambhavs-MacBook-Pro ~ %
```

Mac 3

```
(base) sambhavkumar@Sambhavs-MacBook-Pro ~ % /usr/bin/curl -i
http://localhost:3001/api/status
HTTP/1.0 200 OK
Server: TeamBackend/1.0 Python/3.12.4
Date: Sat, 03 Oct 2026 06:03:53 GMT
X-Backend: A
Cache-Control: no-store
Content-Type: application/json
Content-Length: 76

{"backend": "A", "status": "ok", "time": "2026-10-03T06:03:53.213716+00:00"}% (base)
sambhavkumar@Sambhavs-MacBook-Pro ~ %
```

Mac 2

```
atanuadhikari@Atanus-MacBook-Pro ~ % curl -i http://10.7.7.111:3001/api/status
HTTP/1.0 200 OK
Server: TeamBackend/1.0 Python/3.12.4
Date: Sat, 03 Oct 2026 06:04:38 GMT
X-Backend: A
Cache-Control: no-store
Content-Type: application/json
Content-Length: 76

{"backend": "A", "status": "ok", "time": "2026-10-03T06:04:38.879141+00:00"}%
atanuadhikari@Atanus-MacBook-Pro ~ % curl -i http://10.7.7.111:3002/api/status
HTTP/1.0 200 OK
Server: TeamBackend/1.0 Python/3.12.4
Date: Sat, 03 Oct 2026 06:04:46 GMT
X-Backend: B
Cache-Control: no-store
Content-Type: application/json
Content-Length: 76

{"backend": "B", "status": "ok", "time": "2026-10-03T06:04:46.842678+00:00"}%
atanuadhikari@Atanus-MacBook-Pro ~ %
```

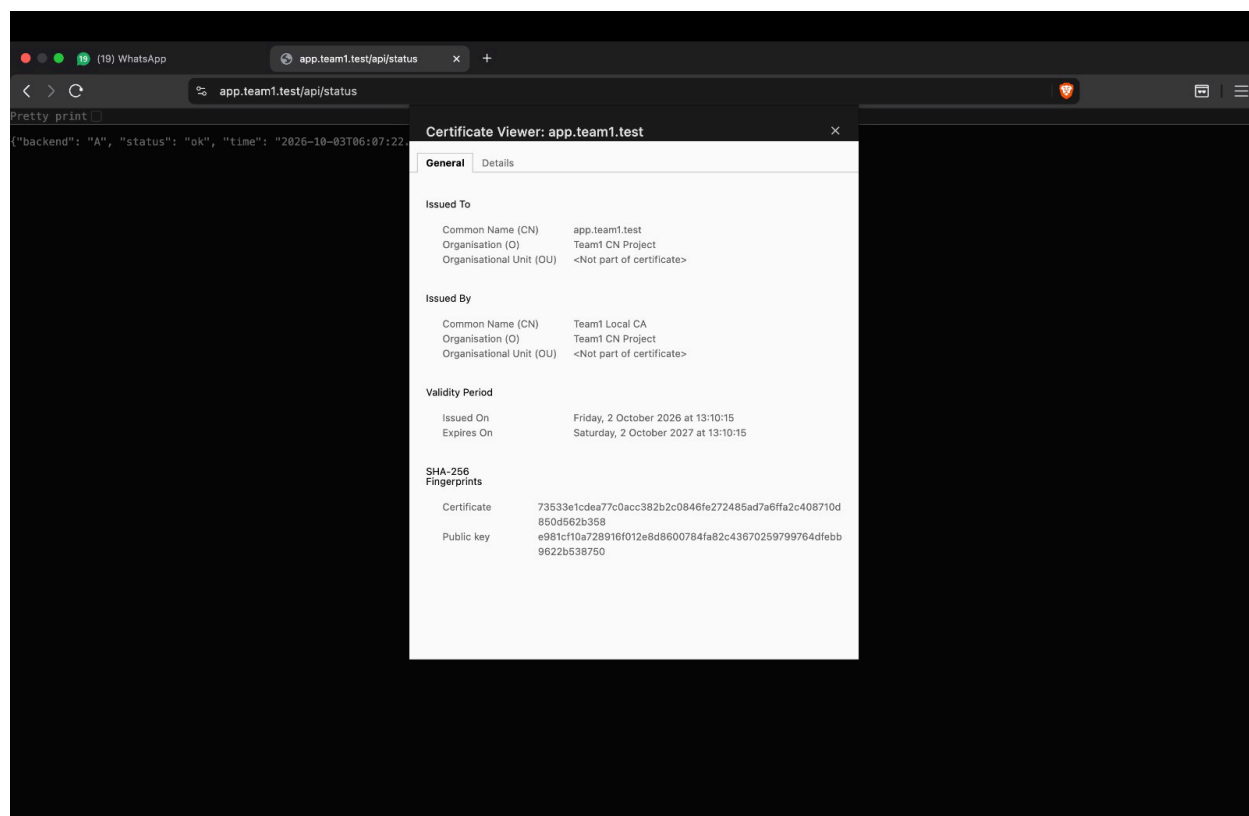
Mac 1

```
princekumarsingh@Princes-MacBook-Pro-3 ~ % curl -sI --http1.1 https://app.team1.test/  
HTTP/1.1 200 OK  
Server: nginx/1.31.6  
Date: Sat, 03 Oct 2026 06:05:54 GMT  
Content-Type: application/json  
Content-Length: 79  
Connection: keep-alive  
X-Backend: A  
Cache-Control: no-store
```

```
princekumarsingh@Princes-MacBook-Pro-3 ~ %
```

```
princekumarsingh@Princes-MacBook-Pro-3 ~ % curl -sI --http2 https://app.team1.test/  
HTTP/2 200  
server: nginx/1.31.6  
date: Sat, 03 Oct 2026 06:06:31 GMT  
content-type: application/json  
content-length: 79  
x-backend: B  
cache-control: no-store
```

```
princekumarsingh@Princes-MacBook-Pro-3 ~ %
```



phase1_full_flow.pcapng

frame.number==240

No.	Time	Source	Destination	Protocol	Length	Info
240	101.213510	10.7.28.177	10.7.7.111	DNS	90	Standard query response 0x647b A app.team1.test A 10.7.

User Datagram Protocol, Src Port: 53, Dst Port: 50413

Domain Name System (response)

Transaction ID: 0x647b

Flags: 0x8580 Standard query response, No error

Questions: 1

Answer RRs: 1

Authority RRs: 0

Additional RRs: 0

Queries

Answers

- app.team1.test: type A, class IN, addr 10.7.16.45
 - Name: app.team1.test
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)
 - Time to live: 60 (1 minute)
 - Data length: 4
 - Address: 10.7.16.45

[Request In: 239]

[Time: 10.265000 milliseconds]

0000 10 9f 41 c2 3a 02 be 7c 69 90 c7 82 08 00 45 00 ..A:..|

0010 00 4c cf a7 00 00 40 11 72 cc 0a 07 1c b1 0a 07 .L...@.

0020 07 6f 00 35 c4 27 00 38 14 b1 64 7b 85 80 00 01 .o.5.'8

0030 00 01 00 00 00 00 03 61 70 70 05 74 65 61 6d 31a

0040 04 74 65 73 74 00 00 01 00 01 c0 0c 00 01 00 01 .test...

0050 00 00 00 3c 00 04 0a 07 10 2d ...<...

Number of answers in packet (dns.count.answers), 2 bytes

Packets: 683 - Displayed: 1 (0.1%)

Profile: Default